

2. (1)

$$\int \frac{x}{1+x^4} dx$$

$$u = x^2 \quad du = 2x dx$$

$$= \int \frac{\frac{1}{2}}{1+u^2} du$$

$$= \frac{1}{2} \arctan u = \frac{1}{2} \arctan x^2$$

(2) $\int \cos^{-1} x dx$

$$\sin^{-1} \cos^{-1}$$

$$= \int \frac{\cos x}{\cos^2 x} dx$$

$$t = \sin x \\ dt = \cos x dx$$

$$= \int \frac{1}{1-t^2} dt$$

$$(1-t^2) = (1-t)(1+t)$$

$$\frac{1}{1-t^2} = \frac{A}{1-t} + \frac{B}{1+t}$$

$$= \frac{A(1+t) + B(1-t)}{(1-t)(1+t)}$$

$$= \frac{(A-B)t + A+B}{1-t^2}$$

$$\Rightarrow \begin{cases} A - B = 0 \\ A + B = 1 \end{cases} \Rightarrow A = B = \frac{1}{2}$$

$$\Rightarrow \frac{1}{1-t^2} = \frac{1}{2} \frac{1}{1-t} + \frac{1}{2} \cdot \frac{1}{1+t}$$

$$= \frac{1}{2} \left(\frac{1}{1-t} + \frac{1}{1+t} \right)$$

$$= \int \frac{1}{2} \left(\frac{1}{1-t} + \frac{1}{1+t} \right) dt$$

$$= \frac{1}{2} \left(-\ln(1-t) + \ln(1+t) \right)$$

$$= \frac{1}{2} \ln \frac{1+t}{1-t} + C$$

$$= \frac{1}{2} \ln \left(\frac{\sin x + 1}{1 - \sin x} \right) + C$$

$$(3) \int \sin^3 x \cos^2 x \, dx$$

$$= \int \frac{\sin^3 x \cdot \cos^2 x \cdot \sin x}{\sin x} \, dx$$

$$= \int \sin^2 x \cos^2 x \cdot \sin x \, dx$$

$t = \cos x$
 $dt = -\sin x \, dx$

$$= \int (1-t^2) \cdot t^2 \cdot (-dt)$$

$$= \int (t^4 - t^2) \, dt$$

$$= \frac{1}{5} t^5 - \frac{1}{3} t^3 + C$$

$$= \frac{1}{5} \cos^5 x - \frac{1}{3} \cos^3 x + C$$

$$(4) \int \sin^2 x \cos^3 x \, dx$$

$$= \int \sin^2 x \cos^2 x (\cos x \, dx)$$

$t = \sin x$

$$= \int t^2 (1-t^2) \, dt$$

$$= \frac{1}{3} t^3 - \frac{1}{5} t^5 + C$$

$$= \frac{1}{3} \sin^3 x - \frac{1}{5} \sin^5 x + C$$

$$(5) \int \frac{\sin^2 x (\cos x dx)}{\cos^2 x}$$

$$t = \sin x$$

$$dt = \cos x dx$$

$$= \int \frac{t^2}{1-t^2} dt$$

$$= \int \frac{t^2 - 1 + 1}{1-t^2} dt$$

$$= \int \left(-1 + \frac{1}{1-t^2} \right) dt$$

$$= -t + \frac{1}{2} \ln \frac{1+t}{1-t} + C$$

$$= -\sin x + \frac{1}{2} \ln \frac{1 + \sin x}{1 - \sin x} + C$$

$$\int \frac{\sin^2 x}{\cos x} dx$$

$$= \int \frac{1 - \cos^2 x}{\cos x} dx$$

$$= \int \left(\frac{1}{\cos x} - \cos x \right) dx$$

c) $\int \sin^2 x \cos^2 x dx.$

$$= \int \frac{1 - \cos 2x}{2} \cdot \frac{1 + \cos 2x}{2} dx$$

$$= \int \frac{1 - \cos^2 2x}{4} dx$$

$$= \int \frac{\sin^2 2x}{4} dx$$

$$= \int \frac{\frac{1}{2}(1 - \cos 4x)}{4} dx$$

$$= \frac{1}{8} \int (1 - \cos 4x) dx$$

$$= \frac{1}{8} x - \frac{1}{32} \sin 4x + C$$

46) $t = \tan x$ $dt = \frac{1}{\cos^2 x} dx$

$$\int \sin^2 x \cos^4 x \cdot \left(\frac{1}{\cos^2 x} dx \right)$$

$$= \int \frac{\sin^2 x \cos^4 x}{(\sin^2 x + \cos^2 x)^3} \cdot \boxed{\frac{dx}{\cos^2 x}}$$

同除 $\cos^6 x$

$$\int \frac{\frac{\sin^2 x}{\cos^2 x}}{\left(\frac{\sin^2 x}{\cos^2 x} + 1\right)^3} \cdot \frac{dx}{\cos^2 x}$$

$$= \int \frac{t^2}{(1+t^2)^3} dt$$

$$187. \int \frac{1}{1-x^2} dx = \frac{1}{2} \ln \frac{1+x}{1-x} + C$$

$$191. \int \frac{\frac{1}{a^2}}{1 - \left(\frac{x}{a}\right)^2} dx$$

$$t = \frac{x}{a}$$

$$dt = \frac{1}{a} dx$$

$$= \int \frac{\frac{1}{a} \cdot \frac{1}{a} dx}{1 - \left(\frac{x}{a}\right)^2}$$

$$\frac{1}{a^2 - x^2} = \frac{1}{2a} \left(\frac{1}{a-x} + \frac{1}{a+x} \right)$$

$$= \frac{1}{a} \int \frac{dt}{1-t^2}$$

$$= \frac{1}{2a} \ln \frac{1+t}{1-t} + C$$

$$= \frac{1}{2a} \ln \frac{1 + \frac{x}{a}}{1 - \frac{x}{a}}$$

(10) $\arctan x + C$

(11) $\int \frac{\frac{1}{a} \cdot \frac{1}{a} dx}{1 + \left(\frac{x}{a}\right)^2} \quad \left(t = \frac{x}{a}\right)$

$$= \int \frac{\frac{1}{a} dt}{1+t^2}$$

$$= \frac{1}{a} \arctan t + C$$

$$= \frac{1}{a} \arctan \frac{x}{a} + C$$

$$(12) \arcsin x + C$$

$$(13) \arcsin \frac{x}{a} + C$$

$$(14) \int \sqrt{1-x^2} dx$$

$$x = \cos t$$

$$dx = -\sin t dt$$

$$= \int \sqrt{1 - \cos^2 t} \cdot (-\sin t) dt$$

$$= \int -\sin^2 t dt$$

$$= \int -\frac{1 - \cos 2t}{2} dt$$

$$= \int \frac{\cos 2t - 1}{2} dt$$

$$= \frac{1}{4} \sin 2t - \frac{1}{2} t + C$$

$$= \frac{1}{2} \sin t \cos t - \frac{1}{2} t + C$$

$$= \frac{1}{2} \sqrt{1-x^2} \cdot x - \frac{1}{2} \arccos x.$$

$$\frac{1}{2} \sqrt{1-x^2} x + \frac{1}{2} \arcsin x + C + C$$

$$\arcsin x = \frac{\pi}{2} - \arccos x$$

$$(15) \int a \sqrt{1 - \left(\frac{x}{a}\right)^2} dx$$

$$= \int a^2 \cdot \sqrt{1 - \left(\frac{x}{a}\right)^2} \frac{1}{a} dx$$

$$t = \frac{x}{a}$$

$$= a^2 \int \sqrt{1-t^2} dt$$

$$= a^2 \cdot \left(\frac{1}{2} \sqrt{1-t^2} \cdot t - \frac{1}{2} \arcsin t \right) + C$$

$$= \frac{1}{2} a^2 \sqrt{1 - \frac{x^2}{a^2}} \cdot \frac{x}{a} - \frac{a^2}{2} \arcsin \frac{x}{a} + C$$

$$\text{(16) } \dot{x} = \sinh t$$

$$\cosh^2 x - \sinh^2 x = 1$$

$$dx = \cosh t \, dt$$

$$\int \cosh t \cdot \cosh t \, dt$$

$$= \int \cosh^2 t \, dt$$

$$= \int \frac{1 + \cosh 2t}{2} dt$$

$$= \frac{1}{2} t + \frac{1}{4} \sinh 2t + C$$

$$= \frac{1}{2} t + \frac{1}{2} \sinh t \cosh t + C$$

$$= \frac{1}{2} \operatorname{arcsinh} x + \frac{1}{2} x \cdot \sqrt{1+x^2} + C$$

$$\int \sqrt{1+x^2} dx \quad x = \tan \theta$$

$$dx = \frac{1}{\cos^2 \theta} d\theta$$

$$= \int \frac{1}{\cos^3 \theta} d\theta \quad \cos^{-3} \theta$$

$$\textcircled{1} \int \frac{\cos \theta}{\cos^4 \theta} d\theta \quad t = \sin \theta$$

$$= \int \frac{dt}{(1-t^2)^2}$$

$$(1-t^2) = (1-t)(1+t) \quad (1-t^2)^2 = (1-t)^2(1+t)^2$$

$$\frac{1}{(1-t^2)^2} = \frac{A}{1-t} + \frac{B}{(1-t)^2}$$

$$+ \frac{C}{1+t} + \frac{D}{(1+t)^2}$$

$$= \frac{A(1-t)(1+t)^2 + B(1+t)^2 + C(1-t)^2(1+t) + D(1-t)^2}{(1-t)^2(1+t)^2}$$

$$= \frac{A(1-t^2)(1+t) + B(1+t)^2 + C(1-t)^2(1+t) + D(1-t)^2}{(1-t)^2(1+t)^2}$$

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$$\textcircled{2} \int \frac{1}{\cos^3 \theta} d\theta$$

$$= \int \frac{1}{\cos \theta} \cdot \frac{1}{\cos^2 \theta} d\theta$$

$$u = \frac{1}{\cos \theta} \quad v = \tan \theta$$

$$dv = \frac{1}{\cos^2 \theta} d\theta$$

$$= \int u \, dv$$

$$u = \sec \theta$$

$$du = \sec \theta \cdot \tan \theta \, d\theta$$

$$= uv - \int v \, du$$

$$= \sec \theta \cdot \tan \theta - \int \tan \theta \cdot \sec \theta \tan \theta \, d\theta$$

$$= \sec \theta \cdot \tan \theta - \int (\sec^2 \theta - 1) \sec \theta \, d\theta$$

$$= \sec \theta \cdot \tan \theta - \int (\sec^3 \theta - \sec \theta) \, d\theta$$

$$\int \sec^3 \theta \, d\theta$$

$$= \sec \theta \tan \theta - \int \sec^3 \theta \, d\theta + \int \sec \theta \, d\theta$$

$$\Rightarrow 2 \int \sec^3 \theta \, d\theta = \sec \theta \tan \theta + \int \sec \theta \, d\theta$$

$$\Rightarrow \int \sec^3 \theta d\theta = \frac{1}{2} (\sec \theta \tan \theta + \int \sec \theta d\theta).$$

$$= \frac{1}{2} \left(\sec \theta \tan \theta + \frac{1}{2} \ln \frac{1 + \sin \theta}{1 - \sin \theta} \right) + C$$

$$x = \tan \theta$$

$$\sec \theta = \sqrt{1 + x^2}$$

$$1 = \sin^2 \theta + \cos^2 \theta$$

$$= \sin^2 \theta + \frac{\sin^2 \theta}{\tan^2 \theta}$$

$$= \sin^2 \theta \left(1 + \frac{1}{\tan^2 \theta} \right)$$

$$\Rightarrow \sin \theta = \sqrt{\frac{1}{1 + \frac{1}{\tan^2 \theta}}}$$

$$= \sqrt{\frac{1}{1 + \frac{1}{x^2}}}$$

$$= \sqrt{\frac{x^2}{x^2 + 1}}$$

$$\cosh^2 x - \sinh^2 x = 1$$

$$\sqrt{x^2 - 1}$$

$$\sqrt{1 + x^2}$$

$$(\sinh x)' = \cosh$$

$$(\cosh)' = \sinh$$

$$\left\{ \begin{array}{l} \cosh 2x = 1 + 2\sinh^2 x = \cosh^2 x + \sinh^2 x \\ \quad \quad \quad = 2\cosh^2 x - 1. \\ \sinh 2x = 2\sinh x \cosh x \end{array} \right.$$

$$(\tanh x)' = \frac{1}{\cosh^2 x}$$

$$(18) \quad x = \tanh \theta$$

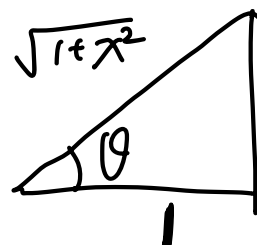
$$\int \frac{1}{\frac{1}{\cosh \theta}} \cdot \frac{1}{\cosh^2 \theta} d\theta$$

$$= \int \frac{1}{\cosh \theta} d\theta = -$$

$$= \frac{1}{2} \ln \frac{1 + \sinh \theta}{1 - \sinh \theta}$$

$\tanh \theta \rightarrow \sinh \theta$

$$= \frac{1}{2} \ln \left(\frac{1 + \frac{x}{\sqrt{1+x^2}}}{1 - \frac{x}{\sqrt{1+x^2}}} \right)$$



$$\sin \theta = \frac{x}{\sqrt{1+x^2}}$$

(20). $x = \sin \theta$

$$\frac{\sqrt{a^2 - x^2}}{x = a \sin \theta}$$

$$\int \frac{\sqrt{4 - 4 \sin^2 \theta}}{2 \sin \theta} \quad dx = 2 \cos \theta \quad d\theta$$

$$= 2 \int \frac{\cos^2 \theta}{\sin \theta} \quad d\theta$$

$$= 2 \int \frac{\cos^2 \theta}{\sin^2 \theta} \quad \sin \theta \quad d\theta$$

$$\underline{t = \cos \theta}$$

$$= 2 \int \frac{t^2}{1-t^2} \cdot (-dt)$$

$$= 2 \int \frac{-t^2}{1-t^2} dt$$

$$= 2 \int \frac{1-t^2-1}{1-t^2} dt$$

$$= 2 \int \left(1 - \frac{1}{1-t^2}\right) dt$$

$$= \text{---}$$

$$= 2t - \ln \frac{1+t}{1-t} + C$$

$$= 2\sqrt{1-\frac{x^2}{4}} - \ln \frac{1+\sqrt{1-\frac{x^2}{4}}}{1-\sqrt{1-\frac{x^2}{4}}} + C$$

$$= \sqrt{4-x^2} - \ln \frac{2+\sqrt{4-x^2}}{2-\sqrt{4-x^2}} + C$$

(21) . $t = 1+2x$

$$dt = 2dx$$

$$\int \sqrt{t} \cdot \frac{1}{2} dt = \frac{1}{3} t^{\frac{3}{2}} + C$$

$$= \frac{1}{3} (\sqrt{1+2x})^3 + C$$

(22) $t = x^2 \quad dt = 2x dx$

$$\int \frac{1}{2} \sqrt{2t+7} dt$$

$$\underline{u = 2t+7 = 2x^2+7} \quad du = 2 dx$$

$$\int \frac{1}{4} \sqrt{u} du$$

$$= \frac{1}{6} u^{\frac{3}{2}} + C$$

$$= \frac{1}{6} (2x^2+7)^{\frac{3}{2}} + C$$

$$(23) \quad t = x^{\frac{2}{3}} \quad dt = \frac{2}{3} \sqrt{x} dx$$

$$\int (2t+1)^{\frac{2}{3}} \frac{2}{3} dt$$

$$u = 2t+1$$

$$(24) \quad x = \sin t$$

$$\int \frac{2 \sin t - 1}{\cos t} \cdot \cos t dt$$

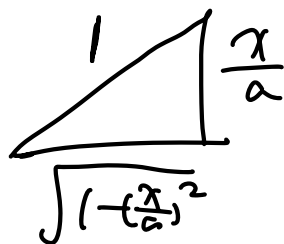
$$= \int (2 \sin t - 1) dt$$

$$= -2 \cos t - t + C$$

$$= \underline{-2\sqrt{1-x^2} - \arcsin x + C}$$

Q 5) . $x = a \sin t$ $\frac{x}{a} = \sin t$

$$\int \frac{1}{(a^2 - a^2 \sin^2 t)^{\frac{3}{2}}} a \cos t \, dt$$



$$= \int \frac{a \cos t}{a^3 (\sqrt{1 - \sin^2 t})^3} dt$$

$$= \int \frac{1}{a^2} \frac{\cos t}{\cos^3 t} dt$$

$$= \int \frac{1}{a^2} \frac{1}{\cos^2 t} dt$$

$$= \frac{1}{a^2} \tan t + C$$

$$= \frac{1}{a^2} \cdot \frac{\frac{x}{a}}{\sqrt{1 - (\frac{x}{a})^2}} + C$$

$$= \frac{1}{a^2} \frac{x}{\sqrt{a^2 - x^2}} + C$$

$$(26) \quad t = x^4 \quad dt = 4x^3 dx$$

$$\int \frac{\frac{1}{4} dt}{\sqrt{1-t^2}}$$

$$= \frac{1}{4} \arcsin t$$

$$= \frac{1}{4} \arcsin x^4.$$

$$Q77). \int \sin x \cdot 2 \sin x \cos x dx$$

$$= 2 \int \sin^2 x \cos x dx$$

$$t = \sin x$$

$$= 2 \int t^2 dt$$

$$= \frac{2}{3} t^3 + C$$

$$= \frac{2}{3} (\sin x)^3 + C$$